

5(a) Higher frequency/lower wavelength means higher energy photons **OR $E = hf$**
[CONCEPT]

violet photons are energetic enough to liberate electrons, while red are not
OR

Energy of violet photons is higher than work function, while red photons is not
[APPLICATION]

Increasing intensity means higher rate of photons [CONCEPT]. Higher rate of photons (incident on potassium leads to) more electrons produced for violet light
[CONCLUSION].

Since red light photons are not energetic enough to liberate electrons, even at very high intensity it will still not emit any electrons from the metal surface [CONCLUSION].

(b) Work function, $\phi = hf_o$ where f_o is the threshold frequency

$$\begin{aligned} f_o &= \frac{\phi}{h} = \frac{4.5 \times 10^{-19}}{6.63 \times 10^{-34}} \\ &= 6.78 \times 10^{14} \\ &= 6.8 \times 10^{14} \text{ J} \end{aligned}$$

'Show' question: Write to more sf than the show answer

So photons below threshold frequency of 6.8×10^{14} Hz will not release electrons from the metal surface.

- (c)(i)** E_k is 0 for f below the threshold frequency 6.8×10^{14} Hz (allow $\pm 0.4 \times 10^{14}$ Hz) because no electrons are emitted from the metal surface.

$E_k = hf - \phi$, so plotting E_k against f gives a graph of constant gradient/straight line is obtained (after threshold frequency).

(ii)
$$h = \text{gradient} = \frac{(5.0 - 0) \times 10^{-18}}{(14.4 - 6.8) \times 10^{14}} = 6.6 \times 10^{-34} \text{ J s}$$

6(c) Binding energy is defined as the amount of energy needed to split a nucleus into its